



CurveMole User Manual

Preview 0.34.0

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CurveMole is a desktop-first, scriptable application for fitting one-dimensional scientific curves. This manual describes the behavior of CurveMole 0.34.0 Preview. The Markdown file is the authoritative source. The LaTeX source and PDF edition are generated automatically from it and must carry the same version number.

Preview status. CurveMole 0.34.0 is suitable for evaluation and controlled testing. It has not yet completed the scientific validation planned for version 1.0.0. Inspect the residuals, parameter correlations, constraints, and exported results before using a fit in research.

Useful links:

- [CurveMole repository](#)
- [Latest release and desktop downloads](#)
- [Issue tracker](#)
- [Version DOI and citation](#)
- [Quick Start](#)

1. About this manual

1.1 Intended audience

This manual is for researchers and technical users who need to fit spectra or other x-y data while retaining control over data preparation, model construction, constraints, uncertainty, and export. Typical data include:

- infrared and Raman spectra;
- powder diffraction patterns;
- kinetic traces;
- chromatographic profiles;
- temperature, pressure, or field-dependent signals;
- any other one-dimensional curve that can be represented by numeric x and y columns.

No programming knowledge is required for the graphical workflow. Separate chapters cover the command-line interface, YAML workflows, Python API, formula builder, and plugins.

1.2 Version policy for the manual

The version in the title and in the line at the top of this file identifies the CurveMole release whose behavior is documented. It is not an independent document version. For example, a manual marked 0.34.0 describes CurveMole 0.34.0.

The release automation performs four checks:

1. the application version and both manual version declarations must agree;
2. all local links in the Markdown source must resolve;
3. Pandoc must generate a standalone LaTeX document without warnings;
4. XeLaTeX must compile the generated source into a valid PDF.

The release assets use versioned names so that a downloaded manual can always be associated with the correct executable.

1.3 Terminology and notation

Menu commands are written as **File > Import data**. Buttons and field names are shown in **bold**. Commands, filenames, identifiers, and parameter paths use monospace text.

CurveMole uses the following residual convention in displays and exports:

$$r_i = y_i - f(x_i).$$

The optimizer minimizes the opposite sign, $f(x_i) - y_i$, after applying uncertainty and spectrum scaling. The sign has no effect on a least-squares minimum but matters when reading exported residuals.

1.4 Scientific responsibility

CurveMole can optimize a mathematically defined model, but it cannot decide whether that model is physically appropriate. A small residual or a high descriptive R^2 does not by itself establish uniqueness, chemical meaning, or predictive validity. In particular:

- do not add components solely to improve a scalar fit statistic;
- use bounds only when they have a defensible meaning;
- inspect correlations and active bounds;
- check whether residuals contain structure;
- report the exact CurveMole version, model, constraints, weighting, and uncertainty method used.

2. Core concepts

CurveMole keeps the scientific engine independent from the graphical interface. The desktop application, command-line interface, YAML workflows, and Python API all operate on the same domain objects.

2.1 Project

A project is the complete analysis workspace. It can contain data, models, masks, transformations, fit results, custom formulas, history, and interface state. Projects are saved as `.fitproj` files.

2.2 Series and curves

A **series** groups related curves. A **curve** contains one x array and one y array, with optional `sigma_x`, `sigma_y`, or point weights. Importing multiple y columns from a common file produces multiple curves in one imported series.

Original numeric arrays are stored as immutable 64-bit floating-point values. Data Calculator operations create reversible transformations instead of overwriting the original arrays.

2.3 Model, component, and parameter

Each curve has its own model. A model is an ordered sequence of components such as a background and several peaks. Every component has:

- a function, for example gaussian or linear;
- a user-facing name;
- a composition operator;
- a set of parameters;
- an enabled or disabled state;
- optional metadata, such as polynomial order or spline x nodes.

A parameter has a current value, optional lower and upper bounds, a fixed state, and an optional expression link. Standard errors and confidence limits are populated after a successful fit when they are available.

2.4 Active, selected, and visible curves

These states are deliberately distinct:

- **Active curve:** the single curve whose model and handles are being edited.
- **Selected curves:** one or more curves chosen for multi-curve operations.
- **Visible curves:** curves drawn in Overlay or Waterfall display modes.

Activating a curve does not automatically select every visible curve. Before a fit, mask transfer, or multi-curve calculation, check the target shown in the relevant dialog.

2.5 Fit plan and result

A fit plan specifies the curves, fit mode, solver settings, per-spectrum weights, and whether numerical contributions should be equalized. A fit result records the optimized parameters, statistics, residual arrays, covariance, correlation, warnings, timestamp, and solver status.

3. Installation and first launch

3.1 Reference requirements

The current reference configuration is:

Item	Minimum or supported target
Processor	x86-64 or Apple Silicon on macOS; 4 CPU cores recommended
Memory	8 GB RAM
Storage	SSD recommended
Display	1920 x 1080 recommended
Windows	Windows 10 22H2 legacy compatibility or Windows 11, 64-bit
macOS	macOS 13 Ventura or newer, Intel or Apple Silicon
Linux	Ubuntu 22.04 LTS, Debian 12, or a comparable newer distribution
Python	Python 3.12 or newer for Python installations

A dedicated GPU is not required. Large multi-curve fits and resampling analyses may require more memory and time than the reference configuration.

3.2 Desktop packages

Open the [latest release](#) and select the package for your system.

Platform	Release asset	Normal launch method
Linux x86-64	CurveMole-VERSION-linux-x86_64.AppImage	Mark executable and run
Windows x86-64	CurveMole-VERSION-windows-x86_64-setup.exe	Run the per-user installer
Windows portable	CurveMole-VERSION-windows-x86_64-portable.zip	Extract and run CurveMole.exe
macOS Apple Silicon	CurveMole-VERSION-macos-arm64.dmg	Drag CurveMole to Applications
macOS Intel	CurveMole-VERSION-macos-x86_64.dmg	Drag CurveMole to Applications

Windows and macOS packages are currently unsigned. The operating system may show a warning on first launch. Download only from the official GitHub release page and verify the checksum if the application will be used for research.

3.3 Linux AppImage

In a terminal opened in the download directory:

```
chmod +x CurveMole-0.34.0-linux-x86_64.AppImage
./CurveMole-0.34.0-linux-x86_64.AppImage
```

If the system cannot mount AppImages through FUSE, use extraction mode:

```
APPIMAGE_EXTRACT_AND_RUN=1 ./CurveMole-0.34.0-linux-x86_64.AppImage
```

Some minimal Linux installations may need graphical runtime libraries supplied by the distribution. On Debian or Ubuntu, the commonly required packages are:

```
sudo apt install libegl1 libgl1 libxkbcommon-x11-0 libxcb-cursor0
```

Choose **Tools > Integrate CurveMole with the desktop** to copy a new AppImage to a stable per-user location, add CurveMole to the application menu, install its icon, and register .fitproj projects. No administrator privileges are required. If a working manual curvemole.desktop launcher already exists, CurveMole backs it up once and adopts its command rather than creating a duplicate application entry.

The Windows installer similarly creates a Start-menu entry, offers an optional desktop shortcut, and registers .fitproj. On macOS the application appears in Applications and Launchpad. On every platform, a registered project can be opened by double-clicking it or choosing **Open with CurveMole**.

3.4 Verifying SHA-256 checksums

Every release includes SHA256SUMS.txt. On Linux:

```
sha256sum -c SHA256SUMS.txt --ignore-missing
```

On macOS:

```
shasum -a 256 CurveMole-0.34.0-macos-arm64.dmg
```

On Windows PowerShell:

```
Get-FileHash .\CurveMole-0.34.0-windows-x86_64-setup.exe -Algorithm SHA256
```

Compare the reported value with the corresponding line in SHA256SUMS.txt.

3.5 Installing the Python package

Use an isolated environment. From a downloaded wheel:

```
python3 -m venv ~/.venv/curvemole
source ~/.venv/curvemole/bin/activate
python -m pip install ./curvemole-0.34.0-py3-none-any.whl
curvemole gui
```

On Windows PowerShell, activation is:

```
py -m venv $env:USERPROFILE\.venv\curvemole
& $env:USERPROFILE\.venv\curvemole\Scripts\Activate.ps1
python -m pip install .\curvemole-0.34.0-py3-none-any.whl
curvemole gui
```

3.6 Running from a source checkout

Clone the repository and install its locked development environment:

```
git clone https://github.com/SebRoLENS/curvemole.git
cd curvemole
uv sync --locked --group dev
uv run curvemole gui
```

For an editable installation using pip:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -e .
curvemole gui
```

3.7 Confirming the installed version

```
curvemole --version
```

The About dialog also displays the version. A project stores the application version that created it, and exports include the application version in machine-readable metadata.

3.8 Version badge and updates

The status bar contains a clickable version badge. CurveMole checks the latest GitHub release at startup and once per hour while the application remains open. The badge is:

- green when the installed release is current;
- yellow when a bug-fix update (x.x.y) is available;
- red when a feature (x.y.x) or major (y.x.x) update is available;
- grey while checking or when the check cannot be completed.

An automatic notification is shown only once for each newly detected release. Click the badge or choose **Help > Check for updates** to check again, inspect the release notes, or reopen the update dialog.

Linux AppImage and Windows desktop packages can normally use **Update now**. CurveMole downloads the matching release asset and verifies the SHA-256 digest supplied by GitHub. Linux replaces the AppImage, Windows portable installations update from the portable ZIP, and installed Windows applications launch the verified installer. The application directory must be writable and no fit or uncertainty task may be running. Existing Windows portable users are offered a one-time migration to the installer; projects, settings and plugins remain in their existing user directories. Python, source, macOS, unsupported architectures, and protected installations must be updated manually from the release page.

4. A complete first fit

This tutorial uses `examples/gaussian.csv` from the source archive. It contains `x`, `y`, and `sigma_y` columns.

4.1 Import the example

1. Start CurveMole.
2. Choose **File > Import data** or press **Ctrl+I**.
3. Select `examples/gaussian.csv`.
4. In the mapping dialog, confirm comma delimiter, decimal point, and header.
5. Select `x` as **X column** and check `y` under **Y column(s)**.
6. Select `sigma_y` as the uncertainty type and column.
7. Press **OK**.

The curve appears in the left tree and becomes active. The central plot displays the data, and the model panel on the right identifies the active curve.

4.2 Add a background

1. Press the **+** button in **Model and parameters**, or press **Ctrl++**.
2. Select **Constant** and enable **Add as background**.
3. Keep composition set to add.
4. Press **OK**.
5. In the parameter table, enter an initial offset near the visual baseline.

4.3 Add a Gaussian peak graphically

1. Add another component and select **Gaussian**. Leave **Add as background** and **Initialize from manually selected points on the graph** unchecked.
2. Press **OK**. The pointer changes to a crosshair.
3. Click at the approximate peak center.
4. Keep the mouse button pressed and drag horizontally from the center to either half-height edge.
5. Release the mouse button.

CurveMole uses twice the dragged distance as the initial FWHM. It estimates a positive peak height from the data minus the existing model, converts this height to the area parameterization, and adds the component as one undoable action.

If placement is incorrect, press **Ctrl+Z** and repeat. Press **Esc** during placement to cancel without adding the component.

4.4 Adjust the model before fitting

Select the Gaussian in the component list. Three green handles appear:

- drag the central target to change center and height;
- drag either vertical side handle to change FWHM;
- edit exact values or bounds in the parameter table.

For this example, a positive lower bound on sigma already exists intrinsically. You may enter a tighter scientifically motivated range, but it is not required.

4.5 Mask an interval if needed

Mask editing is active only while the graphical **Mask** toolbar button is enabled:

1. click **Mask**; the button becomes grey while masking is active;
2. in Single view it acts on the displayed spectrum; in Overlay/Waterfall choose the active spectrum or all visible spectra in the scope prompt;
3. right-drag to mask an interval, left-drag to restore it;
4. click **Mask** again to return to normal navigation.

The example does not normally require masking.

4.6 Run the fit

1. Press **F5** or choose **Fit > Fit**.
2. Use **Single / independent** mode.
3. Check that the example curve is enabled in the table.
4. Keep **Local least squares (automatic)** and linear loss.
5. Press **OK**.

After a successful fit, CurveMole commits the optimized parameter values back into the displayed model and immediately redraws the model sum, individual components, and residual panel. Parameter standard errors appear in the ± 1 sigma column when covariance is available. Open **View > Diagnostics** to inspect residual warnings.

4.7 Save and export

Save the complete workspace with **File > Save project as**. Use a .fitproj extension.

Then choose **File > Export analysis bundle**. Select a destination and choose both how to save and **What to export?**. The default selection writes only fit_results.csv, which associates every curve with its model functions and lists each parameter together with its fitted value and standard error. Data tables, plots, reports, JSON, reusable models, project copies, uncertainty matrices, and diagnostics are optional.

5. Main window and controls

5.1 Series and curves tree

The left dock contains a search field and a tree with **Visible**, **Series / Curve**, and **State** columns.

- Click a curve to activate it.
- When switching spectra, the function selected in Model & Parameters stays at the same position in the function list, regardless of its name or type. If the new spectrum has fewer functions, its last function is selected; an empty model clears the parameter table. Show all functions retains its project-wide selection.
- Use Ctrl-click or Shift-click for a multi-selection, or use **Select all** and **Deselect all** above the tree.
- Use **Remove selected** to delete one or more accidentally imported curves. Removal is undoable.
- Use **New series** to create an empty named series. Empty series are retained when the project is saved.
- Right-click one or more selected spectra and choose **Move selected to series** to reorganize them.
- Right-click selected spectra and choose **Move selected up/down** to change their order inside a series. Multiple adjacent spectra move as a stable group.
- Right-click a series and choose **Merge series into** to append all its spectra to another series and remove the source series.
- Moving, merging, creating, and reordering series are all Undo/Redo operations and do not invalidate fitted parameters.
- Use the visibility checkbox to include or exclude it from Overlay and Waterfall displays. Hover over the checkbox for an explanation. Hiding a spectrum retains its data and fit; hidden spectra are excluded from fit plans. Highlighting rows separately selects spectra for operations.
- Double-click an editable name, or right-click and choose **Rename data...** or **Rename series....** Series names must remain unique.
- Right-click a series and choose **Delete series...** to remove it together with all its curves, models and results. A populated series requires confirmation; Undo restores the complete series. Original files on disk are retained.
- Right-click a **curve** and choose **Choose data colour...** to set its colour. Red is reserved for the fitted Model sum and cannot be assigned to a spectrum.
- Right-click a **series** and choose **Series palette** to recolour the complete series with one of the built-in non-red palettes. Palette and individual colours are saved in the project.
- Use the search field to filter curves by name without removing them.

Curve states are:

State	Meaning
Not fitted	No successful fit is associated with the current curve state
Ready	The curve is prepared for fitting
Running	A fit involving the curve is in progress
Fitted	The last fit completed successfully
Modified/outdated	Data or model changed after the last successful fit
Failed	The latest fit attempt failed

5.2 Plot workspace

The central workspace contains the data/model plot and an optional residual plot. Controls above it provide:

- **Display:** Single, Overlay, or Waterfall;
- **All series / Active series / Selected:** one-click scope for Overlay and Waterfall;
- **X offset** and **Y offset:** display-only Waterfall spacing;
- **Residuals:** show or hide the linked residual panel;
- **Autoscale:** automatically frame the experimental data when the active spectrum changes;
- **All points / Unmasked points:** choose whether automatic framing includes masked data;
- **Plot style:** draw imported data as Lines, Points, or Lines + points;
- **Visual only - background-subtracted:** preview subtraction of marked background functions without changing data or fit inputs.

Choose **View > Plot appearance...** for separate data, function and residual styles, line widths, symbols, colours and grid settings. These display settings are stored with the project.

With Autoscale off, switching spectra preserves the view. **View all** and **View unmasked** explicitly frame the corresponding experimental points; model extrapolations, labels and handles do not affect these limits.

The main toolbar contains the graphical **Mask** button. Its scope is chosen on entry in Overlay/Waterfall; mouse buttons determine the operation.

Offsets never modify data and are removed when converting pointer positions back to scientific coordinates. Fitting and export use the original numeric coordinates, not screen offsets.

The coordinate readout displays the pointer coordinate and the nearest finite point from the active curve. Overlay and Waterfall views use adaptive peak-preserving downsampling and clip drawing to the visible range for dense spectra. Single view keeps the full display resolution. Descending x arrays may be reversed only in the temporary display copy so that fast clipping remains available. Fitting, project storage, and export always use the complete original arrays.

Model functions initially receive names such as **Voigt1**, **Voigt2**, and **Gaussian1**. Right-click a function and choose **Rename function...** to assign any non-empty name, including Unicode. The name belongs to that component of that spectrum; internal identifiers and parameter links do not change. Renaming supports Undo/Redo, preserves fitted status, and survives saving/reopening, duplication and copying. Plot labels, notebook entries, uncertainty tables and exports use the name. Their labels are shown above each function maximum by default and are shifted vertically when necessary to avoid overlap. Right-click the main plot and toggle **Show component labels** to hide or show these labels. After every completed fit, CurveMole explicitly applies the returned optimized parameters to the displayed model and redraws the plot while preserving the current zoom. The **Model sum is always red**; imported spectra and selectable series palettes exclude red so a data curve can never be confused with the fitted sum.

5.3 Navigation and axes

Normal plot navigation is available when neither placement nor Mask mode is active. The mouse wheel remains available during Quick Add peak placement, spline placement, Quick Fit, and live fit refreshes. A refresh preserves the current viewport instead of resetting a zoom chosen by the user. When the plot has keyboard focus, Up and Down activate the previous or next spectrum in project order, matching navigation in the curve tree.

Use **View > Axes** for:

- automatic range;
- logarithmic x or y;
- reversed x or y;
- locked plot view.

Reversing an axis is only a view setting. Logarithmic axes do not transform the data used by the fit. Values incompatible with a log display may not be visible even though they remain stored.

5.4 Model and parameters dock

The right dock normally lists components for the active curve. Enable **Show all functions** to list every model function in the project, grouped by series and spectrum. Group headings cannot be selected as functions. The list supports Ctrl-click and Shift-click multi-selection, including selections across spectra. Mathematical composition order matters because operators are evaluated sequentially. **Reorder** sorts only this displayed list and automatic names; it preserves the composition order and fit. The Up/Down controls explicitly change composition order.

Select all selects every displayed function. **Select all except backgrounds** selects only functions that are not marked as background and clears any previous background selection. In Show all functions mode these actions span the project. Selection is separate from enabling: a checked function contributes to the model and fit; an unchecked function remains stored but is excluded. Hover directly over its checkbox for an explanation. Changing a checkbox on a selected function applies the enabled state to the whole selection. **Mark as background** classifies the selected functions; it does not subtract them from the data.

The controls below the list are:

Control	Action
+	Add a component
Duplicate	Duplicate the selected component and its settings
Up / Down	Change component order
Delete	Delete the component after confirmation
Reorder	Sort the function list and renumber automatic names using a parameter value
Advanced reorder rules	Choose the numbering parameter separately for each function type
Copy fit	Copy selected model information to other curves
Copy fit to next	Open the copy dialog with the next curve in the source series selected
Copy parameter	Copy one parameter from one source function to chosen project-wide targets

Bulk actions on selected functions can delete, duplicate, reorder, enable or disable, mark or unmark backgrounds, and fix or free their parameters as applicable. Each bulk edit is one Undo/Redo operation. Uncheck a component in the list to disable it without deleting it. Disabled components are not evaluated or fitted.

Copy parameter is enabled only when exactly one source function is selected. Select a row in the parameter table to make that parameter the default, open the copy window, and explicitly check the target functions. The window shows the source spectrum, function, parameter, and numeric value. By default only the value is copied; optional controls

also copy bounds, fixed/free state, and the parameter link/relation. Targets without the parameter or whose retained bounds reject the source value are skipped and reported. The complete copy is undoable.

5.5 Quick actions

The main toolbar includes **Quick Add Function** and an adjacent function selector. The selector contains the complete current registry: built-ins, project formulas, reusable user formulas, and loaded plugin functions. CurveMole remembers the last selection across sessions and also updates it when a function is added through the ordinary Add dialog.

Quick Add uses the insertion choice last confirmed for that function in Add component. Linear and Cubic spline default to manual points; normal peaks use centre/FWHM click-drag placement. Functions whose point initialization is off are otherwise added immediately. Click Quick Add again to add another function. Section 8 describes point placement, initial locking and graphical controls.

The toolbar also provides the most common project, model, fitting, background, and calculator actions as icons. Hover over an icon to see its action name.

5.6 Dockable tools

The **View** and **Tools** menus show or hide these docks:

- **Worksheet:** read-only view of current x, y, uncertainty, weight, and mask state;
- **Diagnostics:** residual summary and warnings;
- **Log:** operational messages and errors;
- **Data Calculator:** reversible scalar and curve-to-curve transformations;
- **Function Builder:** safe mathematical custom functions;
- **Uncertainty Analysis:** Monte Carlo, bootstrap, and profile calculations;
- **Laboratory notebook:** project notes and searchable object descriptions;
- plugin panels contributed by installed plugins.

Opening a tool places it in the shared tab group beside **Model and parameters** and selects its tab. Closing or hiding a tab preserves the panel and its current state; opening it again returns to the same panel. Tabs can still be moved or floated. When many tabs are open, use the arrows to scroll through their titles. Titles longer than 25 characters end in an ellipsis; hover over one to read its full title. CurveMole remembers window geometry, dock arrangement, and theme. Use **View > Reset layout** to restore the standard tabbed layout.

5.7 Themes

View > Theme provides System, Light, and Dark themes. Theme choice affects only presentation. It does not change plot data, exported numeric values, or fitting.

5.8 Drag and drop

Drop a .fitproj file onto the window to open it. Drop any other local file to begin content-aware data import, regardless of its extension. If a drop contains both a project and data files, the first project is opened.

6. Importing data

6.1 Supported text formats

The graphical importer shows .txt, .dat, .csv, and .tsv as common choices but can open any text-based file whose contents form a valid numeric table. For example, spectrometer files named .xy, .xye, or with a vendor-specific suffix can be selected through **All files** or dropped onto the window. The extension is not used as proof of validity. Files must contain at least two columns and at least two rows after parsing.

CurveMole detects common delimiters:

- whitespace;
- comma;
- semicolon;
- tab;
- vertical bar.

It also proposes decimal point or decimal comma and whether the first data row is a header. Always inspect the preview before accepting the mapping.

6.2 Comments, blank lines, and encoding

Blank lines are ignored. Lines whose first non-space character is #, ;, %, or ! are treated as comments by the default importer. UTF-8 with an optional byte order mark is the default encoding.

CurveMole also detects leading instrument metadata or preamble lines before the rectangular table. **Ignore first rows** shows the proposed number of physical lines to skip and can be changed from 0 to 1,000,000. Skipping occurs before delimiter, decimal, and header interpretation; therefore a column-name row after the preamble can still be used as the first data-row header. The accepted value is stored in the curve import metadata for reproducibility.

The semicolon can therefore be ambiguous: a line beginning with semicolon is a comment, while semicolons inside table rows can be delimiters. Check the preview for locale-specific exports.

Before confirming files, the file chooser already includes a small spectrum graph. Highlight a file to preview it; select X and Y from the adjacent dropdowns (first and second column by default). Reading happens in the background. The graph uses the same peak-preserving adaptive renderer as the main plot and full-data axis limits. Choices are remembered per file and proposed in the subsequent import mapping dialog. With multiple files, Apply this mapping to all files deliberately uses the mapping confirmed for the first file. Files requiring custom parsing can still be opened when the quick preview is unavailable, then configured in the mapping dialog.



Figure 1: Preview while selecting files, before opening the import mapping dialog.

6.3 Column mapping

Choose exactly one x column and one or more y columns. Each selected y column becomes a separate curve sharing the selected x column. Use **Select all Y columns** or **Deselect all Y columns** when importing tables with many signals. The Python API additionally supports explicit x-y column pairs.

Optional columns are:

- `sigma_x`;
- `sigma_y`;
- generic weight;
- variance;
- inverse variance.

Only one of `sigma_y`, generic weight, variance, or inverse variance can be selected for a given import. Variance is converted to `sigma_y` by square root. `sigma_x` is stored in the project but is not used by the version 0.34.0 optimizer.

The spectrum preview below the numeric table updates as you change X/Y columns, uncertainty mappings, delimiter, decimal separator, header or skipped rows. It plots the complete file, reusing the main graph's peak-preserving downsampling and clipping rules. Parsed data are cached while changing columns. Descending X is reversed for display; nonmonotonic X is not clipped/downsampled. The view bounds always come from full-resolution finite X/Y pairs. With no selected Y or an invalid mapping, the preview clears and explains the problem. For batch import, it previews the file named in the mapping dialog.

6.4 Batch import

When importing multiple files, **Apply this mapping to all files in this batch** reuses the first file's parsing and column mapping. Enable it only if all files truly share a layout. CurveMole does not silently guess a new mapping for an incompatible file in the same batch.

Files are imported in natural filename order: `scan_2` precedes `scan_10`, regardless of the order returned by the file chooser.

Calibrated WinSpec SPE files

CurveMole also reads binary WinSpec SPE 2.x files containing calibrated one-dimensional acquisitions. The first column is wavelength in nm, followed by one intensity column per frame. Select the frame through the Y-column mapping. SPE 3.x, uncalibrated axes, images and multiple-ROI acquisitions are unsupported; export those acquisitions to calibrated x-y text before importing.

Automatic folder import

Choose **File > Automatic folder import...** to add arriving files to the current project. Select an existing folder; an optional case-insensitive **Filename contains** filter, the one-based X/Y columns and a **Wait for stable file** delay (2 seconds by default). Monitoring starts only when you press **Start**.

- **Also import matching files already in the folder** includes existing files; otherwise the session starts with newly arriving files.
- **Import changed files again as NEW spectra** retains the previous acquisition and adds a new one when file contents change.

- **Show newest imported spectrum** controls whether successful imports take focus; it can be changed while monitoring runs.
- **Automatic workflow** selects Import only or a loaded plugin's import processor. Configure the plugin first; processing runs on the new acquisition in the background before it is added to the live project.

Only files directly in the folder are scanned. Symlinks, hidden/temporary files and projects are ignored, and automatic inputs are limited to 256 MiB. Identical content at the same path is not imported twice within a session. Each successful acquisition is one Undo operation. Failed files appear in the log; correct them or the settings and choose **Retry failed files**. Imports wait while fitting, a desktop automation or a modal dialog is active. Closing the monitor panel leaves it running; choose **Stop** or **File > Stop automatic import** to stop. Changing projects ends the session, and reopening CurveMole does not start monitoring automatically.

6.5 Invalid numeric cells

Non-numeric cells are converted to missing numeric values. Rows are retained in the stored arrays but excluded from fitting when x or y is not finite. A row is also excluded if:

- `sigma_y` is non-finite or not strictly positive;
- a point weight is non-finite or negative;
- it is masked;
- it lies outside active fit ranges defined through the core API.

CurveMole does not reorder rows, average duplicate x values, or interpolate missing values during import.

6.6 Uncertainties and weights

If absolute `sigma_y` values are supplied, point residuals are scaled as:

$$r_i^{(w)} = \frac{f(x_i) - y_i}{\sigma_{y,i}}.$$

If inverse variances $w_i = 1/\sigma_i^2$ are supplied, CurveMole multiplies residuals by $\sqrt{w_i}$. A generic weight column directly multiplies residuals by its value. This distinction is intentional and is recorded in the curve metadata.

Zero generic weight is allowed and gives a point zero numerical influence. A zero or negative `sigma_y` is invalid and excludes the row.

7. Curves, masks, and transformations

7.1 Immutable originals

CurveMole preserves imported x, y, uncertainty, and weight arrays. Displayed working arrays are reconstructed by applying a transformation history to the originals. This design makes data preparation reversible and auditable.

7.2 Masking points and intervals

Masked points remain stored but are excluded from fitting. They are drawn as faded markers, and masked intervals are shaded.

Enable the graphical **Mask** button: right-drag masks an interval and left-drag unmasks it. A right/left click performs the equivalent operation on a point. The button becomes grey while active. Turn it off to navigate without changing exclusions. Changing display mode also exits masking.

7.3 Mask targets

Target	Effect
Active	Changes only the active curve
All visible	Changes every currently visible curve

Single view always uses the active spectrum. Overlay and Waterfall prompt for Active or All visible when entering Mask mode; cancelling leaves masking off.

For interval masks, the same numeric x interval is applied to every target. For a single-point mask transferred from the active curve, CurveMole locates the nearest x value on other targets and applies it only if it lies within the configured tolerance.

Set this tolerance through **Data > Mask transfer tolerance**. A value of zero requires an exact matching x value on transferred point masks.

7.4 Undo and mask history

A graphical mask action is placed on the application Undo stack. Press **Ctrl+Z** to undo it or **Ctrl+Shift+Z** on platforms that use that standard Redo binding. The project also stores mask arrays and numeric mask intervals.

Fits, including Quick Fit and sequential fitting, also support Undo/Redo. Undo restores the pre-fit models, parameter values, fit results, and sequential pause state; Redo restores the completed fit without running the solver again. Cancelled or failed worker operations restore the previous state. The Undo stack retains the latest 20 operations (fits and other edits combined); input data arrays are not duplicated for fit history. Undo/Redo are disabled while a worker is running.

Quick Fit is available before opening Fit...: its first run uses the default independent-fit settings on the selected curves, falling back to the active curve. Later runs retain the

configured solver, loss and algorithm settings but always use **Single/Independent**, even after a sequential or global fit. Open Fit... to use another fitting mode.

7.5 Data Calculator

Open **Tools > Data Calculator**. The following scalar operations are available:

- add to or subtract from y;
- multiply or divide y;
- shift x;
- scale x;
- normalize y by maximum absolute value;
- normalize y by signed integrated area.

Every calculator operation (including restoring data) offers **Active curve** or **Choose multiple spectra...**. The picker initially shows the active series; **All series** shows every spectrum grouped under colored series headings. Use checkboxes or Select all shown. Hidden spectra are excluded. Cancel changes nothing. A batch is checked in full before any spectrum is modified; one Undo restores the whole batch. Normalization also scales `sigma_y` consistently. Dividing by zero and normalizing a zero or invalid signal are rejected.

Advanced column formulas

The green **Advanced - column formula** entry is first in the operation list, followed by the green **Custom formula** for a single plotted axis. New imports retain every source column as numeric data in the project, including columns not plotted. Original column order and row alignment are preserved. `c1`, `c2`, `c3`, etc. refer to one-based source columns; `x` and `y` refer to the current plotted axes. **Insert into formula** lists the available codes with original column names; choose an entry, then press **Insert** to add its code at the cursor. **Write result to** chooses X, Y or any retained column for the result. The hint below it explains that Apply writes the formula result into this column. Merely selecting an input column does not edit the formula or change its destination. A live summary such as `Y - Counts <- c2 / c3**2` makes the direction explicit. **Spectra to modify** separately chooses which spectra receive the operation.

For a file with Energy, Counts and Monitor columns, plotting `c1` against `c2`:

1. choose **Advanced - column formula**;
2. choose **Y - Counts** as the destination;
3. insert `c2` and `c3` from the dropdown to enter `c2 / c3**2`;
4. choose Active curve or multiple spectra, then Apply non-destructively.

You can also enter `y / c3^2` (^ and ** both mean exponentiation in Advanced), `c2 - c4`, or `sqrt(c5)`. A header may be referenced as `${Monitor counts}`; codes remain unambiguous for unusual names. Only restricted mathematical expressions are accepted, never Python execution. Scalar results are broadcast to every row; missing columns and non-finite output (including division by zero) reject the batch. Masks do not change row alignment: formulas operate on all rows.

Writing to a plotted source column updates that axis. Writing to an unplotted column leaves the graph unchanged and makes the new values available to later formulas. Normal calculator operations on X/Y are also reflected in their mapped columns. Calculations are local to each spectrum, even when spectra originated in the same file. In a multi-spectrum batch `cN` always means position N in each file; check that the files have compatible layouts. Uncertainty propagation is not inferred for custom formulas; changing an auxiliary uncertainty column does not remap the uncertainty arrays originally selected for fitting.

Name and **Save formula** to retain the expression and destination in the project, then recall it through Saved formulas. Retained columns, formulas and their Undo/Redo history

The image shows a dark-themed user interface for an advanced calculator. It features several dropdown menus and text input fields. The 'Operation' dropdown is set to 'Advanced — column formula'. The 'Write result to' dropdown is set to 'Y — Counts'. The 'Formula' text input contains 'c2 / c3**2'. The 'Insert into formula' dropdown is set to 'Choose a column to insert...', with an 'Insert' button next to it. Below this, there is explanatory text about the 'Insert' function and a preview of the formula 'Y — Counts ← c2 / c3**2'. The 'Saved formulas' dropdown is set to '(not saved)'. The 'Formula name' text input contains 'Monitor correction'. At the bottom, there are three buttons: 'Save formula', 'Apply non-destructively', and 'Restore original data'. The 'Spectra to modify' dropdown is set to 'Active curve'.

Operation: Advanced — column formula

Write result to: Y — Counts

The formula will update this column when you apply it.

Formula: $c2 / c3^{**2}$

Insert into formula: Choose a column to insert... Insert

Insert adds a column reference at the cursor. It does not choose where the result is written.
Use c1, c2, ... for imported columns; x and y for current axes. Example: $c2 / c3^{**2}$.

Y — Counts ← $c2 / c3^{2}$**

Saved formulas: (not saved)

Formula name: Monitor correction

Save formula

Spectra to modify: Active curve

Apply non-destructively

Restore original data

Figure 1: Advanced calculator with column insertion and destination controls in dark mode.

survive `.fitproj` saving without the original source file. Projects created before this feature contain only their previously stored axes: re-import the source to recover additional columns.

7.6 Curve-to-curve calculations

The calculator can add, subtract, multiply, or divide by another curve. The operand is aligned to the target x grid by:

- linear interpolation;
- nearest-neighbor interpolation;
- cubic spline interpolation.

Extrapolation is disabled by default. Outside the operand range, aligned values become missing and will therefore be excluded from later fitting. Enabling extrapolation is an explicit advanced choice. Cubic interpolation requires at least four unique finite operand x values.

The aligned operand array is stored with the transformation so that reopening the project does not depend on a later change to the source curve.

7.7 Restoring original data

Restore original data removes all currently applied transformations from the selected target while retaining them on the redo stack. The imported arrays are not re-read from disk. The whole action is also integrated with application Undo.

7.8 Worksheet

The Worksheet is a read-only inspection view. It shows up to 100,000 rows for the active curve with columns for x, y, `sigma_y`, weight, and mask state. It is not a spreadsheet editor and does not modify source arrays.

8. Building models

All entries in the function library are fitting functions. Constant, linear, polynomial, cubic-spline, peak-shaped, and custom formulas are not separated into a special background-function class. Background is instead a property of a model component. Select a component in **Model and parameters** and enable **Mark as background** when that component represents the experimental baseline or background.

Data > Subtract background... uses these component-level marks. If no component is marked, CurveMole first asks which model functions should be designated as background. If background components already exist, it asks which of the marked components should be subtracted. The subtraction uses the current resolved/fitted parameter values, is reversible with Undo, applies over the complete data array, and disables the subtracted components afterwards to prevent double-counting.

The same dialog can apply the operation to every eligible spectrum, using the enabled functions already marked as background in each model. **Data > Revert background...** lists only spectra changed by this dedicated subtraction command and restores the selected ones, including the relevant function states. **Visual only - background-subtracted** instead removes the enabled marked background only from the plot; it never changes curve arrays, fit inputs, or exported values.

During graphical cubic-spline placement, nodes may be placed anywhere in plot coordinates, including outside the x/y extent of the measured data. Adding nodes does not auto-range the graph. Left-drag continues to pan and the mouse wheel continues to zoom while placement is active.

8.1 Component composition

Components are evaluated in list order. Available operators are:

Operator	Operation
add	Add the component array to the accumulated model
subtract	Subtract the component array
multiply	Multiply the accumulated model by the component array
divide	Divide the accumulated model by the component array
convolve	Convolve using FFT and scale by median x spacing

The first enabled component can use only add or subtract; multiplication, division, or convolution requires an existing accumulated model. Additive peak plus background models should normally use add for every component.

Division by a component that reaches zero and other non-finite model evaluations are rejected by the fitter.

8.2 Built-in peak parameterization

Every built-in peak uses signed integrated area, not peak height, as its intensity parameter. Negative areas produce negative peaks. This common parameterization makes areas comparable between different line shapes.

Gaussian

Parameters: area, center, sigma.

$$G(x) = \frac{A}{\sigma\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x - x_0}{\sigma} \right)^2 \right].$$

The derived FWHM is 2.354820045σ . sigma is intrinsically positive.

Lorentzian

Parameters: area, center, gamma.

$$L(x) = \frac{A\gamma}{\pi[(x - x_0)^2 + \gamma^2]}.$$

gamma is the half width at half maximum, so FWHM is 2γ . gamma is intrinsically positive.

Voigt

Parameters: area, center, sigma, gamma.

The Voigt component is the area-normalized convolution of a Gaussian with standard deviation sigma and a Lorentzian with HWHM gamma. CurveMole evaluates it with the SciPy Voigt profile. Both width parameters are intrinsically positive.

The displayed FWHM is the approximation:

$$\text{FWHM} \approx 0.5346(2\gamma) + \sqrt{0.2166(2\gamma)^2 + (2.354820045 \sigma)^2}.$$

Pseudo-Voigt

Parameters: area, center, fwhm, eta.

The profile is an area-normalized linear mixture with a common FWHM:

$$pV(x) = A[(1 - \eta)G_{\text{unit}}(x) + \eta L_{\text{unit}}(x)].$$

fwhm is intrinsically positive and eta is intrinsically bounded to $[0, 1]$. eta=0 is Gaussian and eta=1 is Lorentzian.

8.3 Built-in functions commonly used as backgrounds

The library names are **Constant**, **Linear**, **Polynomial** and **Cubic spline**. They become background components only when **Add as background** in the Add dialog or **Mark as background** in the model panel is enabled.

Constant

Parameter: offset.

$$B(x) = c.$$

Linear

Parameters: intercept, slope.

$$B(x) = c_0 + c_1x.$$

Polynomial

Choose an order from 0 to 50 when adding the component. Parameters are c_0 , c_1 , and so on through the chosen order:

$$B(x) = \sum_{k=0}^n c_k x^k.$$

High polynomial orders are numerically sensitive when x values are large or poorly scaled. Prefer a low order or rescale x with the Data Calculator when scientifically appropriate.

Cubic spline

Spline x nodes are metadata, rather than optimizer parameters. Their y values (y_0 , y_1 , and so on) are normal parameters that may be fitted, fixed, bounded, or linked. Two nodes produce linear interpolation. Three or more nodes produce a natural cubic spline, with extrapolation outside the node range.

New graphically placed spline y nodes are fixed by default so an intentionally drawn baseline does not move when fitting begins. Uncheck **Fixed** for individual nodes, or use **Lock all** / **Unlock all** below the parameter table for every parameter in the selected function. The spline remains continuous through masked intervals even though masked measurements do not contribute to the objective function.

8.4 Quick Add Function

Choose a registered function from the selector beside **Quick Add Function** in the main toolbar. The list includes built-in peaks and backgrounds, generic functions, project-contained formulas, reusable formulas, and trusted plugin functions. The last choice is remembered.

- a peak with manual-point initialization off opens centre/FWHM placement and finishes after one click-drag; click Quick Add again for another peak;
- a cubic spline normally opens point-by-point node placement;
- Linear normally opens manual-point placement; other functions can use it when enabled in the Add dialog;
- functions with point initialization off are added with their initial parameters; a spline in this mode uses an initial node grid with free y parameters.

Quick Add therefore uses the same registry and initialization logic as the normal Add dialog. Selecting a function in the normal Add dialog also makes it the next Quick Add choice. The manual-point and background options chosen in Add component are also remembered for Quick Add separately for each function.

8.5 Adding a peak with the pointer

After choosing a peak and pressing **OK**:

1. click the intended center;
2. drag horizontally to either half-height edge;
3. release to set the initial full width.

A green region previews the width. If no horizontal width is selected, CurveMole uses a default based on 5 percent of the x span, local point spacing, and floating point precision.

For built-in peaks, CurveMole converts the graphical height and FWHM into the native area and width parameters. New custom peaks should declare explicit semantic roles for their position, intensity, and width parameters in Function Builder. Older custom functions without role metadata retain compatibility through conventional names such as center, x0, fwhm, sigma, gamma, area, amplitude, or height.

8.6 Initializing any function from manually selected points

In Add component, enable **Initialize from manually selected points on the graph**. This is selected by default for Linear and Cubic spline and can be used for other built-in, custom or plugin functions. For a custom formula, Function Builder can set **Use manual-point initialization by default**.

Click explicit points representing the function you want to create. A live green function preview updates as points are added or removed; it is an initial estimate, not a fit to the measured spectrum. Linear requires at least two points. Other non-spline functions require at least two points and at least as many points as parameters. Cubic spline requires at least two distinct x positions.

Use **Finish** or double-click to commit after the minimum number of points; **Undo point** removes the most recent point, right-click removes the nearest point, and **Esc** cancels insertion. The resulting parameters are fixed by default. Use **Unlock all** or individual **Fixed** checkboxes before optimizing them. Points and the corresponding controls are stored with the component and project.

Drawing a cubic-spline background

After choosing **Cubic spline**, enable **Add as background** and keep manual-point initialization enabled:

1. left-click background nodes on the plot;
2. inspect the live green preview, which updates after each click;
3. use **Undo point** if necessary;
4. finish with **Finish** or double-click after at least two nodes; right-click removes the nearest node;
5. press **Esc** to cancel the entire placement.

Nodes are sorted by x. Clicking again at effectively the same x updates that node instead of creating a duplicate. Exact duplicate x positions are not permitted.

8.7 Graphical editing after placement

Select a peak component to display its handles. Drag the central target to change center and area-derived height. Drag a side line to change FWHM. For a Voigt peak, changing FWHM graphically scales sigma and gamma by the same ratio, preserving their current relative contribution.

For a function created from manual points, selecting it displays its original control points. Unlock the relevant parameters before dragging a point in x and y. For a spline, unlock that node's y parameter: dragging also updates its stored x position and keeps nodes sorted. For other functions, at least one parameter must be free; CurveMole refines those free values against the moved control points while retaining fixed and linked values. Each completed drag is undoable. Exact numeric parameter edits are available in the table.

For normal peak centre/width handles, fixed values cannot normally be changed by a drag. Hold **Ctrl** while dragging to change the stored value while leaving the parameter fixed. Manual-point handles instead require unlocking the relevant parameters. Linked values cannot be dragged independently because an expression controls them.

8.8 Automatic peak suggestions

Choose **Model > Find positive peaks....** Despite the menu label, the first dialog can search positive, negative, or both signs. The next dialog selects any registered function classified as a peak, initially preferring the current Quick Add peak. CurveMole then:

- subtracts the median as an initial baseline;
- estimates noise through a robust median absolute deviation;
- uses a default prominence of the larger of three noise estimates or 1 percent of the y range;
- estimates width at half prominence;
- sorts suggestions by prominence.

Choose how many suggestions to add. Every suggestion is initialized using the selected peak function's explicit parameter roles or, for legacy functions, recognized parameter names. Suggestions are initial estimates, not a scientific decision about the number, identity, or line shape of peaks.

8.9 Duplicating, disabling, deleting, and reordering

Duplicate a component to create an adjacent copy with a new internal identifier. Disable a component to compare models without losing its settings. Deletion and reordering are undoable. Reordering can materially change models that use operations other than addition or subtraction.

The **Reorder** button sorts the function list in ascending parameter order (peak center, or X position, by default), including peaks of different types. It also renumbers automatically named functions within each type to match that order. IDs, links and custom names are preserved; custom-named functions are sorted too. Functions without a finite value for their sorting parameter keep their list positions, and equal values retain their relative order. The adjacent **Advanced reorder rules...** button selects another sorting and numbering parameter for each function type and remembers those rules. Both the list order and automatic names are restored by Undo and reapplied by Redo. **Reorder changes presentation only:** parameter values, links, fit results and the mathematical composition order remain unchanged, including multiplication, division and convolution. The display order is saved with the project and copied with the model. Use the Up/Down controls to change the mathematical composition order explicitly; those controls restore the list to composition order.

8.10 Copying a fit to other curves

Model > Copy fit copies selected information from the active curve. Use **Select all** or **Deselect all** when choosing many target curves. The dialog initially shows **Active series**; **All series** exposes the entire project, grouped under colored series headings. Only targets in the displayed scope are copied.

Copy fit to next, beside Copy fit, opens the same options with the next spectrum in the source series already checked. It is unavailable at the end of that series. Copying remains undoable. Options include:

- component structure;
- current or best values;
- bounds and fixed states;
- internal links;
- background components;
- masks;
- fit ranges.

Masks and fit ranges are excluded by default because x grids and excluded regions may not correspond scientifically. When copying masks, point transfer uses the configured x tolerance. Review every target model after copying.

9. Parameters, constraints, and links

9.1 Parameter table

The table columns are:

Column	Meaning
Parameter	Parameter name; linked parameters show a link indicator
Value	Current value or successful fitted value
± 1 sigma	Covariance-based standard error when available
Analysis - / +	Separate lower/upper distances from the saved uncertainty analysis selected for this spectrum
Fixed	Exclude the parameter from optimization
Lower	Lower bound; blank means negative infinity
Upper	Upper bound; blank means positive infinity
Link	Graphical Set link... control; linked parameters show their source in readable form

Press **Set link...** to choose a source spectrum, component, and parameter. The default relationship is **Equal to source**. Choose **Advanced expression** for relationships such as $2 * \text{\$}\{\text{source}\} + 1$. Use **Remove link** to make the parameter independent again.

Source spectrum also determines what happens when you copy the functions:

Source spectrum choice	After copying to another spectrum
This spectrum (follows copies) (default)	Use the corresponding source function in the destination spectrum
Specific spectrum: spectrum name	Keep referring to that exact spectrum, even if it is the current spectrum

For example, link Gaussian2's center to Gaussian1's center in Spectrum 1. After copying the functions to Spectrum 2, **This spectrum** links Gaussian2 to Gaussian1 in Spectrum 2. **Specific spectrum: Spectrum 1** links Gaussian2 in Spectrum 2 back to Gaussian1 in Spectrum 1. The dialog shows the source and explains the copy behavior as you change the selection. This choice is saved with the parameter and retained by Undo/Redo, parameter copying and sequential model propagation. Removing the link resets the choice to **This spectrum**.

An edit is validated immediately. CurveMole rejects a value outside its bounds, a lower bound above an upper bound, an invalid expression, a missing referenced parameter, a link cycle, or a linked value that violates its own bounds.

9.2 Parameter states

A parameter is exactly one of:

- free;
- fixed;
- lower-bounded;
- upper-bounded;
- interval-bounded;
- linked.

Intrinsic function bounds always remain active. For example, clearing the visible lower cell cannot make a Gaussian sigma negative because the function definition sets a positive intrinsic minimum.

9.3 Parameter paths

The canonical path is:

```
curve_id.component_id.parameter_name
```

A link refers to a path inside `${...}`. For example:

```
${curve_ab12.component_cd34.center}
```

or a derived relation:

```
${curve_ab12.component_cd34.center} + 12.5
```

The graphical **Set link...** dialog creates these paths automatically, so normal GUI use does not require knowing curve or component identifiers. The canonical expression is still shown as a tooltip and remains available to advanced workflows, Python code, and machine-readable exports. For a link between different spectra, select both spectra and use **Global simultaneous** mode; CurveMole blocks incompatible fit modes with an explicit message.

9.4 Link evaluation

Links form a directed dependency graph. CurveMole resolves the graph before fitting, detects cycles, and evaluates linked values at each model evaluation. Linked parameters are not independent optimizer variables. If covariance is available, their standard errors are propagated numerically from free parameters.

9.5 Safe expression language

Formulas and links are parsed as a restricted expression tree. They are never passed to unrestricted Python `eval` or `exec`.

Allowed arithmetic includes `+`, `-`, `*`, `/`, power `**`, remainder `%`, unary signs, comparisons, boolean combinations, and conditional expressions. Constants are `pi`, `e`, and `inf`.

Allowed functions are:

```
abs, sqrt, exp, log, log10,  
sin, cos, tan, arcsin, arccos, arctan,  
sinh, cosh, tanh,  
erf, erfc,  
minimum, maximum, clip, where, heaviside
```

Attribute access, imports, indexing, comprehensions, arbitrary function calls, file access, and network access are not part of the expression language.

10. Fitting

10.1 Pre-fit checklist

Before pressing **F5**:

1. confirm the active and selected curves; use **Select all** / **Deselect all** when appropriate;
2. inspect invalid and masked points;
3. choose a physically defensible background and peak count;
4. provide reasonable initial values;
5. add justified bounds for poorly determined parameters;
6. inspect links for cycles or unintended dependencies;
7. check that each selected model contains at least one free parameter.

10.2 Fit modes

Single / independent

Each selected curve is fitted as an independent problem. Results are then combined for reporting. Parameter links between different independently fitted curves cannot act as a simultaneous cross-curve constraint; use Global mode for that purpose.

Sequential

Sequential mode is a propagating refinement. Choose an **Initial source spectrum** whose model and parameters have already been inspected and approved. The source is not re-fitted. Its model is cloned to the next selected spectrum, fitted there, and the result becomes the source for the following spectrum. Selected spectra before the chosen source are excluded; target spectra do not need models in advance.

Use **Functions excluded from copying** to select any source functions to leave out. Functions marked as background are excluded by default; uncheck one to copy it. Functions already present on each target retain their values, metadata, constraints, and order during copying; the fit still respects their fixed/free state. A function with the same internal identity is kept rather than duplicated. Distinct functions remain distinct even if their names match. Target-local functions are not carried into later spectra. The selection is also preserved when resuming a paused sequence. This is independent of ignoring functions in the parameter-change pause monitor.

For newly copied functions, parameter values and structure are propagated. The dialog separately controls whether to preserve bounds, fixed/free state, internal links, background tags, enabled/disabled state, and composition/grouping. **This spectrum** links are remapped to corresponding target parameters; **Specific spectrum** links keep their original references. Cross-spectrum links require Global simultaneous fitting rather than independent or sequential fitting. Structural metadata required by the function, such as spline nodes and custom-formula metadata, is always retained.

Two optional safeguards can pause a sequence:

- normalized residual RMSE worsens by both a selected factor and a minimum fraction of the signal scale; defaults are 2.5 times and 2 percent;

- a free parameter changes by more than a selected normalized amount; the default is 75 percent.

Specific source functions can be excluded from the parameter-change trigger while remaining fully copied, fitted, and included in the residual. If fitting or a safeguard pauses the sequence, the affected spectrum becomes active and a persistent **Continue sequential fit** button appears in the status bar. Inspect or edit that spectrum, then continue; the accepted current model becomes the new propagation source. CurveMole does not silently skip a paused spectrum. The warning explains the affected spectrum, reason, measured change and threshold, and next action. **Last pause reason**, beside the sequential controls, reopens the latest warning while the sequence is paused, including after a manual fit used to correct the paused spectrum. The button disappears when the sequence completes or is terminated. The message is saved with the project and is replaced when a later sequential fit pauses.

Global simultaneous

All selected residual arrays are concatenated into one optimization problem. Models may differ between curves. Parameter links can connect curves and express shared, offset, scaled, or otherwise related values.

10.3 Per-spectrum weights

The Fit dialog uses the same **Active series / All series** scope for independent, sequential and global fits, with colored series headings. Hidden spectra are excluded from the plan; the sequential source chooser follows the same scope. Weights and checkbox choices are retained while switching scope. Function selection and enabled/background checkboxes are described in Section 5.4.

The Fit dialog accepts a positive numeric spectrum weight for every selected curve. If s_j is the spectrum weight, all point residuals from spectrum j are multiplied by $\sqrt{s_j}$.

Scale each spectrum to equal numerical contribution additionally divides the residual vector of each curve by the square root of its number of usable points. This prevents a densely sampled curve from dominating only because it contains more rows. It does not normalize signal amplitude or uncertainty and is never enabled silently.

10.4 Local least squares

The default is local nonlinear least squares using SciPy. CurveMole automatically selects:

- Levenberg-Marquardt for an unbounded ordinary least-squares problem;
- trust-region reflective when bounds or a robust loss require it.

The graphical dialog exposes the maximum number of evaluations and confidence level. New fit plans default to 1,000 evaluations and use a step tolerance that permits a successful early stop when parameter changes become numerically negligible. During local optimization, the status progress is updated and the plotted trial model is refreshed every 20 evaluations. These refreshes preserve the user's current zoom and do not replace the stored parameters unless the fit completes successfully. Advanced algorithm options, the Python API and YAML settings expose tolerances, scaling, local method, random seed and Differential Evolution controls. The solver menu also offers explicit TRF, Dogbox, and LM, plus Nelder-Mead, Powell, and L-BFGS-B. The latter three minimize the selected loss directly; they can be useful for difficult models but may take more evaluations. LM needs unbounded parameters and linear loss. Hover over a solver to see its intended use.

10.4.1 Advanced algorithm options and Reset default

The default remains **Local least squares (automatic)** with **linear** loss. **Advanced algorithm options** shows only the controls relevant to the chosen solver. Numeric fields accept scientific notation such as 1e-8; tooltips explain each option.

Algorithm	Configurable controls
Automatic, TRF, Dogbox, LM	ftol, xtol, gtol, x_scale; automatic mode also offers the local method
Differential Evolution	Generations, population multiplier, mutation minimum/maximum, recombination, relative/absolute tolerance, seed and automatic search limits; least-squares controls apply to local refinement
Nelder-Mead	Iteration limit, xtol, fatol, adaptive coefficients and optional initial simplex
Powell	Iteration limit, xtol, ftol
L-BFGS-B	Iteration limit, ftol, projected-gradient tolerance, line-search steps and correction history

Maximum evaluations remains configurable for all built-in solvers. For Differential Evolution it limits the local refinement; generations/population control the global search. Numerical Jacobian probes and L-BFGS-B numerical gradients may require additional model calls beyond the solver's reported evaluation budget.

An iteration limit of 0 retains the minimizer's automatic limit. An optional Nelder-Mead simplex is a JSON matrix with N+1 rows and N columns in the fitter's free-parameter order (selected curve, component and parameter order, including referenced link dependencies). Leave it blank for normal automatic initialization. Vertices must satisfy bounds. LM requires unbounded parameters and linear loss.

Reset default restores all algorithm/loss parameter fields, evaluation budget and confidence level. It preserves the selected algorithm and loss, fit mode, spectrum selection/weights and model parameters. Defaults include 1,000 evaluations, ftol=1e-10, xtol=1e-8, gtol=1e-10, x_scale=jac, and loss scale 1. DE defaults are 400 generations, population multiplier 15, mutation 0.5-1, recombination 0.7, relative tolerance 0.01, absolute tolerance 0 and seed 1729. Nelder-Mead and Powell tolerances default to 1e-4; L-BFGS-B uses ftol=2.220446049250313e-9, gtol=1e-5, 20 line-search steps and 10 corrections. These are starting configurations, not guarantees for every dataset.

10.5 Robust losses

Available losses are:

- linear for ordinary least squares;
- soft_l1, which smoothly weakens the effect of large residuals;
- huber, squared near zero and nearly linear for large residuals;
- cauchy, which strongly suppresses very large residuals.

Hover over the loss options in the Fit dialog for short descriptions. Losses alter the penalty assigned to residuals, not the mathematical peak shape.

Robust loss can reduce the influence of large residuals, but it is not a substitute for understanding outliers, detector artifacts, or an incomplete model. AIC, AICc, and BIC are reported only for the linear loss. Robust-loss covariance uses a sandwich estimate, and CurveMole recommends explicit resampling.

For `soft_l1`, `huber` and `cauchy`, the **Loss scale (f_scale)** field appears. It must be positive and sets the transition between small and large residuals. It is hidden for `linear`, where it has no effect. The scale is expressed in the residual units passed to the optimizer: signal units without uncertainty weighting, standardized residual units with `sigma_y`, further modified by spectrum weights and equal-contribution scaling. The default of 1 is not an automatic noise estimate; choose a scale appropriate to the data. Reset default restores it to 1.

10.6 Differential Evolution initial search

Choose **Differential Evolution + local refinement** when a local initial estimate is not sufficient. Differential Evolution searches globally, then passes its best point to the local least-squares solver.

When a free parameter lacks a finite bound, CurveMole creates one around its current value. By default, the search extends 50% of the absolute current value below and 50% above it. At zero, a scale based on the plotted data is used. Edit the separate **Below** and **Above** percentages in the Fit dialog; existing explicit bounds always take precedence. These temporary search limits do not change the parameter table or the saved model. Inspect them conceptually before interpreting the result: a global search cannot discover a solution outside its search interval. It is usually much slower than a well-initialized local fit.

10.7 Cancellation and failed fits

Use **Fit > Cancel running task** to request cancellation. The request is checked during residual evaluation and resampling. The window cannot close while a background task remains active.

If the solver raises an error or does not converge, CurveMole restores the last valid parameter values instead of committing an incomplete optimizer vector. The curve is marked Failed, the error is written to the Log, and the last successful result remains separate from the failed attempt.

10.8 Common causes of failure

- no free parameters;
- fewer than two usable points;
- initial value outside bounds;
- non-finite model values;
- division by zero in a composed model;
- a missing or cyclic link;
- a linked value outside its bounds;
- an invalid or overly narrow Differential Evolution search interval;
- an underdetermined model;
- inadequate initial values or maximum evaluations.

11. Results, statistics, and diagnostics

11.1 Visual results

The main plot shows data, total model, individual component curves, and residuals. The parameter table shows fitted values and standard errors. A curve state changes to Fitted only after solver success.

The full numeric result is stored in the project and written to exports. Preview 0.34.0 does not yet provide a single comprehensive on-screen results table, so use the analysis bundle for archival inspection.

11.2 Reported global statistics

Let N be the number of usable observations and k the number of independent free parameters. CurveMole reports:

Statistic	Definition or interpretation
Degrees of freedom	$N - k$
RSS	$\sum_i (y_i - f_i)^2$ without point weighting
RMSE	$\sqrt{\text{RSS}/N}$
Chi-square	Sum of squared weighted residuals
Reduced chi-square	Chi-square divided by $N - k$ when positive
Descriptive R^2	$1 - \text{RSS}/\text{TSS}$
AIC	Gaussian ordinary least-squares expression when applicable
AICc	Small-sample correction when $N > k + 1$
BIC	Gaussian ordinary least-squares expression when applicable

The descriptive R^2 can be misleading for baselines, constrained fits, or models without an intercept. It is included as a familiar descriptor, not a universal model quality score.

11.3 Per-curve statistics

Each curve output records the original usable indices, x , observed y , fitted y , residual, weighted residual, N , RSS, RMSE, and descriptive R^2 . The global result combines all curves in the plan.

11.4 Covariance and correlation

For ordinary least squares, covariance is derived from the fit Jacobian. If all curves provide `sigma_y`, the uncertainty is treated as absolute by default and the covariance is not rescaled by residual variance. Otherwise residual variance scales the covariance.

CurveMole uses a pseudo-inverse when the Jacobian is rank deficient and reports a warning. Covariance is unavailable when degrees of freedom are not positive. A warning is also generated when at least one free-parameter correlation has absolute value 0.95 or greater.

Symmetric normal intervals are clipped to active bounds. A parameter at a bound is flagged because its true uncertainty is generally asymmetric and not adequately summarized by a symmetric covariance interval.

11.5 Residual diagnostics

Open **View > Diagnostics** after a fit. The summary reports:

- Durbin-Watson statistic;
- number of residuals with absolute standardized value at least 3;
- lag-1 autocorrelation;
- warnings for potential outliers and autocorrelation beyond an approximate 95 percent white-noise band.

The analysis bundle exports the autocorrelation sequence for every fitted curve. A structured residual pattern usually indicates that the model, background, weighting, or data treatment deserves review.

12. Explicit uncertainty analyses

Open **Fit > Uncertainty Analysis** only after a successful fit. These calculations run in the background, can be cancelled, and may be substantially slower than the baseline fit. **Fit covariance** displays the uncertainty already calculated by the fit without running replicas. Monte Carlo, residual bootstrap and block bootstrap default to **200 replicates** (not optimizer iterations), and remain configurable. The optimizer's default evaluation budget is unchanged. Two hundred replicas are a quick estimate; use more and check stability when precise percentile endpoints matter.

CPU processes sets the maximum number of independent spectra analyzed in parallel, with remaining spectra queued. The default is up to four processes, limited by the available CPU count. Profile scans can parallelize their grid points; analyses involving executable plugin functions use one process. Raising the process count can increase memory use, and it does not change the requested replicate count.

Run on selects the active spectrum, spectra selected in the project tree, or all fitted spectra. Each independent spectrum uses its own last successful fit, even if another spectrum was fitted more recently. Switching the active spectrum updates the report shown in the panel. Results are saved separately for each spectrum and method. **Displayed uncertainty** selects which saved analysis supplies the coloured **Analysis – / +** column in **Model and parameters** for the active spectrum. The most recently completed resampling/profile analysis is selected automatically; the choice is saved per spectrum and can be changed or set to **Fit error only**. The adjacent **±1σ** column always retains the original fit error. The analysis column shows the distances below and above the fitted value separately. It shows a dash when a parameter has no valid interval or the interval excludes the original fitted value; inspect the assessment for details. The parameter CSV retains parameter_err for the fit error and adds parameter_analysis_err_minus, parameter_analysis_err_plus, and the selected method when applicable. In **File > Export functions parameters, Also export covariance matrices** is off by default. When selected, a sibling *_covariance folder contains an index.csv and labelled matrices: the fit covariance and, separately, sample covariance for the selected bootstrap or Monte Carlo analysis when raw samples are available. Global fits retain cross-spectrum terms in one joint matrix. Profile likelihood provides intervals, not a sample covariance matrix. Running an uncertainty analysis never changes the original fit covariance. Older projects may need a new fit for spectra whose individual baselines were not recorded. Profile likelihood accepts one active spectrum at a time; coupled global fits are analyzed together when resampling is requested.

All methods display a table with spectrum/function names, parameter, original fitted value, confidence interval endpoints and assessment. Click the assessment to read its explicit reasons, including the names and correlation coefficients of strongly correlated parameters. Intervals are not symmetric +/- errors and their midpoint is not necessarily the original fitted value. Technical paths, seed and the baseline timestamp remain available under **Technical details**. The confidence level is configurable for resampling/profile; covariance shows the recorded fit's confidence level. Changing the method recalls its most recent recorded analysis.

Assessment	Meaning
OK	No diagnostic issue is flagged by these checks; this does not establish practical adequacy
Attention	Bound contact, strong correlation, failed replicates, an open profile scan, or another stated caveat

Assessment	Meaning
Critical	No usable interval, too few successful replicas, rank-deficient covariance, or an interval spanning most of the allowed range
Fixed	Parameter was fixed; its uncertainty was not estimated

These are deterministic diagnostic heuristics, not universal scientific thresholds: strong correlation means absolute $r \geq 0.95$; coverage of at least 80% of a finite allowed parameter range is flagged; fewer than 20 successful replicas is critical, and fewer than 200 is flagged for unstable percentile endpoints. Correlations use empirical samples for resampling when available, otherwise the fit covariance. A parameter touching a bound is not necessarily physically wrong. Click the assessment to see why it is flagged; no color certifies the correctness of the physical model. Analyses describe their recorded baseline; editing data or models requires refitting before resampling. Reports remain inspectable but are marked outdated when appropriate.

12.1 Parametric Monte Carlo

This method generates synthetic y values from the fitted model plus independent normal noise using the imported absolute σ_y array. Every selected curve must therefore provide valid σ_y .

Each synthetic data set is refitted using the local solver. The result records the requested, completed, and failed replicates, seed, free parameter paths, samples, empirical intervals, and failure messages.

12.2 Residual bootstrap

Residuals are centered, sampled independently with replacement, added to the fitted model, and refitted. This assumes residual exchangeability and can be inappropriate when residuals are serially correlated or heteroscedastic.

12.3 Block bootstrap

The block bootstrap resamples contiguous circular residual blocks. Enter a block length or choose Automatic. The automatic estimate uses the first lag at which the absolute residual autocorrelation falls below e^{-1} , subject to practical limits.

Block resampling is often preferable for spectra or kinetic traces with local correlation, but the block length remains a scientific modeling decision.

12.4 Profile likelihood

Choose one independent, non-linked parameter. CurveMole fixes it across a finite grid, refits all remaining free parameters, and compares chi-square with the baseline. The default grid spans approximately three covariance standard errors on either side, or a fallback span when no standard error is available, while respecting bounds.

The default uses 31 grid points and a one-parameter chi-square threshold. The grid size, confidence level and lower/upper scan limits can be changed in the panel; blank limits

select the automatic range, and explicit limits must respect parameter bounds. Failed grid points are counted. The desktop profile currently requires a single-spectrum baseline; it does not profile a global multi-spectrum fit. A profile interval is more informative than a symmetric standard error near bounds or in nonlinear problems, but grid resolution should be considered when interpreting endpoints.

12.5 Reproducibility and failed replicates

The default seed is 1729 unless changed through programmatic settings. Resampling outputs record configuration and up to the first 100 failure messages. Do not report an empirical interval without also reporting how many replicates completed and failed. Selecting **Uncertainty intervals, assessments and matrices** in the export dialog also writes `uncertainty/parameter_assessments.csv` for the recorded analyses, including names, fitted values, interval endpoints, confidence level, targets and reasons. Covariance/correlation matrices remain available in the same export option.

13. Projects, saving, and recovery

13.1 The .fitproj format

A .fitproj is a ZIP container with:

- a versioned JSON manifest;
- original arrays stored as NumPy .npy payloads;
- uncertainty and weight arrays when present;
- mask arrays;
- transformation operands;
- models, links, custom functions, results, history, and UI/export state;
- SHA-256 checksums for every binary payload.

NumPy arrays are loaded with `allow_pickle=False`. CurveMole projects never require executing a Python pickle. The format is documented in [project-format.md](#).

13.2 Atomic saves

CurveMole writes a temporary sibling file, validates the archive, and then atomically replaces the destination. A failed save should therefore not partially overwrite a previous valid project.

13.3 Project locks and read-only mode

Opening a project creates a sibling .lock file containing process and host information. If another instance already holds the lock, CurveMole opens the project read-only. Use **Save project as** to create an editable copy. A stale lock left by an abnormal termination may need manual inspection and removal after confirming that no other instance is using the project.

13.4 Normal save and portable copy

Save project updates the current project path. **Save portable copy** writes a complete separate .fitproj without changing the active project path. Since project data are already embedded, the portable copy is intended for sharing or archiving a snapshot.

Projects are not encrypted. Use suitable filesystem permissions or encrypted storage for confidential data.

13.5 Autosave recovery

Every ten minutes, a modified revision is written to the operating system's CurveMole user cache. An unchanged revision does not create another recovery. The three newest valid, distinct recovery files for a project are retained.

Only after a crash or abnormal exit, CurveMole offers **Recoverable sessions** at startup. Normal launches do not prompt, even when older deferred copies exist. The same list is available from **File > Recoverable sessions**. Sessions are grouped by project name with

their recovery date and copy count. **Recover** opens the newest valid copy by default; an older retained copy can also be selected. The recovered workspace is unsaved and does not overwrite the original project.

Decide later retains the copies. **Delete recovery copies** removes all backups for the selected session after confirmation. Saving a project normally or explicitly choosing **Discard** when closing or switching projects removes its recovery copies. Cancelling a dialog or a failed save preserves them. Other projects' copies remain.

File > Recent projects lists up to ten successfully opened or saved projects; choose an entry to reopen it, or clear the list from that menu.

13.6 Unsaved changes

An asterisk in the window title marks a dirty project. New Project, Open Project, and Quit ask whether to save, discard, or cancel when unsaved changes exist.

14. Exporting an analysis bundle

14.1 Starting an export

Choose **File > Export analysis bundle** or press **Ctrl+E**. Export mode and export content are independent.

The existing save modes remain:

- leave both mode checkboxes clear to create a new export in the selected directory;
- **Create versioned export** to create a timestamped subdirectory;
- **Update existing CurveMole-owned files after confirmation** to update files that CurveMole can verify as belonging to the current project export.

Under **What to export?**, select the desired outputs. Only **Fit results** is checked by default.

14.2 Default fit-results file

The default export creates only `fit_results.csv`. No additional visible file and no empty subdirectory is created.

Each row represents one parameter of one model component and includes the series, curve and curve ID, component and component ID, displayed function and function ID, enabled/background state, composition operator, parameter name and full parameter path, fitted value, standard error, confidence interval, bounds, fixed/link state, unit, human-readable value with uncertainty, and derived area/FWHM where defined.

This format makes the association between every data curve, the functions used to fit it, and the fitted parameters explicit in a single table.

14.3 Optional outputs

The dialog can additionally export:

- data plus fitted-function wide CSV tables;
- a tidy CSV table for Python;
- machine-readable fit results as JSON;
- a reusable `.fitmodel`;
- a `.fitproj` project copy, optionally including full uncertainty samples;
- the main plot as PNG and/or SVG;
- compact HTML and PDF summaries;
- a full reproducibility HTML report;
- the laboratory notebook as UTF-8 text;
- all recorded settings for the last fit as CSV;
- covariance and correlation matrices when available;
- residual autocorrelation diagnostics when available;
- an export README.

Directories such as `data/`, `python/`, `figures/`, `report/`, `uncertainty/`, and `diagnostics/` are created only when a selected output actually writes a file there. An option whose required fit result is unavailable does not create an empty folder.

14.4 Safe overwrite policy

New exports store their ownership list in the project export configuration rather than creating a hidden sidecar file. This allows the default export to contain truly only `fit_results.csv`. Existing `.curvemole-export.json` manifests from older versions are still read for backwards-compatible updates.

When **Update existing** is selected, CurveMole refuses to overwrite a colliding file unless it can verify that the file belongs to the current project export. Unrelated files in the directory are preserved.

14.5 Wide data tables

Optional wide CSV files are intended for direct inspection and tools such as Origin or QtiPlot. Depending on data and model, columns include:

- x and y;
- total fit;
- residual as data minus fit;
- total background;
- every enabled component;
- `sigma_y` or point weight;
- mask flag.

Rows remain aligned with the complete curve, including masked and invalid entries.

14.5.1 Exporting optimizer and loss settings

Last fit: all solver/loss settings (CSV) writes `fit_settings.csv`, containing all recorded `FitSettings` fields, including advanced solver parameters, loss and `f_scale`. Each row includes the fit timestamp, mode and curve IDs. It describes the selected last fit result, not unsaved edits in the Fit dialog; fields for inactive algorithms are retained for completeness. No settings file is produced without a fit result. `fit_results.csv` separately contains all model parameter values, bounds, links and available errors. The JSON result includes the same optimizer settings; the full reproducibility HTML report also shows them as a table.

14.6 Tidy data and JSON

`python/data_tidy.csv` stores one quantity per row with series, curve, component, x, value, mask, and validity fields. `python/results.json` records schema version, application version, result metadata, parameters, statistics, warnings, and settings without duplicating large numeric arrays.

14.7 Reports and reusable files

- `summary_report.html` is a compact human-readable report.
- `report/full_reproducibility.html` includes project metadata.
- `report/summary.pdf` contains a summary plot and statistics.
- `.fitmodel` stores the first project model and custom formulas for programmatic reuse.
- `.fitproj` is a complete portable project snapshot.

The exported report PDF is an analysis summary, not the CurveMole software manual.

14.8 Exporting one numeric file per spectrum

Choose **File > Export spectra and fit curves...** or press **Ctrl+Shift+E** when the result should be immediately replottable as one multi-column table per spectrum. This is separate from the structured analysis bundle. Choose any project spectra and any of these options:

- export the measured spectrum with its fitted background subtracted;
- omit masked rows;
- include every enabled individual model function;
- include the combined fitted background;
- include the total fit;
- include residuals as data minus total fit.

The x and spectrum columns are always written. Output names follow `<spectrum>_curvemole.<original extension>`; duplicate names receive a numeric suffix. The original delimiter is retained when practical, otherwise the output is tab-delimited. Headers describe axis labels, units, functions, background state, total fit, and residuals, and never contain whitespace so they remain safe for whitespace parsers. An unknown original extension is preserved.

14.9 Exporting function parameters

File > Export functions parameters... writes a wide CSV table with one row per project spectrum and two header rows identifying functions and parameters. Each parameter has value and fit-error columns; functions with the same displayed name align across spectra, with duplicate names disambiguated. Spectra that are not currently Fitted retain their row but have blank numeric cells. Saved selected-analysis errors are added as separate minus/plus columns, together with the method. This is a focused numeric export; use the analysis bundle for the full constraints and reproducibility record. **Also export covariance matrices** optionally writes the labelled sibling matrices described in Section 12.

15. Command-line interface

15.1 General form

```
curvemole [--version] [--json] [--verbose] COMMAND [OPTIONS]
```

Use `--json` for machine-readable output. Normal data go to standard output; logs and errors go to standard error.

15.2 Open the GUI

```
curvemole gui
```

The desktop executable can also receive a project or data path as its first argument.

15.3 List available functions

```
curvemole functions list
curvemole functions list --json
```

15.4 Fit one file

```
curvemole fit spectrum.csv \
  --x wavenumber \
  --y absorbance \
  --function gaussian \
  --background linear \
  --output spectrum.fitproj
```

`--x` and `--y` accept a column name or zero-based index. Optional initial values are `--center`, `--width`, and `--area`. The meaning of `--width` follows the selected function: Gaussian and Voigt use sigma, Lorentzian uses gamma, and pseudo-Voigt uses fwhm.

If no initial peak override is supplied, the CLI uses the strongest automatic positive peak suggestion when available. Background choices are none, constant, or linear.

15.5 Fit a series

```
curvemole fit-series scan_01.csv scan_02.csv scan_03.csv \
  --x 0 --y 1 \
  --function pseudo_voigt \
  --background constant \
  --mode sequential \
  --output series.fitproj
```

Modes are independent, sequential, and global. Use `--global-search` to request Differential Evolution. Missing finite bounds are generated from current parameter values; use a YAML workflow or the Python API to customize percentages or explicit parameter bounds.

15.6 Inspect and validate

```
curvemole inspect analysis.fitproj --json
curvemole inspect raw_data.dat --json
curvemole validate analysis.fitproj
curvemole validate workflow.yml
```

Inspecting a data file reports detected parsing and a preview. Validating a project checks archive structure and checksums. Validating YAML checks the supported workflow schema and required top-level structures.

15.7 Export an existing project

```
curvemole export analysis.fitproj exported_analysis
```

Use `--versioned` for a timestamped subdirectory. Use `--force` only when updating a directory whose CurveMole ownership manifest permits the overwrite.

15.8 Exit codes

Code	Meaning
0	Success
2	Command usage error
3	Data validation error
4	Fit error
5	Project format error
6	Plugin or other CurveMole domain error
10	Unexpected failure

16. Reproducible YAML workflows

16.1 Running a workflow

```
curvemole run examples/gaussian_workflow.yml --json
```

Relative paths are resolved from the workflow file's directory. Schema version 1 supports imports, custom functions, explicitly trusted plugins, models, constraints, fit settings, project output, and bundle export.

Run an existing automation from the desktop

The separate **Automations** toolbar at the bottom contains **Run automation**. On the first click, choose a .yml or .yaml workflow; CurveMole validates and runs it. Later clicks run the same file, including after restarting CurveMole. The tooltip shows its path. Use the arrow beside Run to **Choose automation** or **Stop automation**. Cancelling the file picker leaves the previous choice intact; if the remembered file is missing, Run asks for another file.

The workflow executes in a separate process and uses its own imported spectra, models and output settings. It does not operate on the currently open GUI project. Completion reports the processed spectra, fit convergence and output paths. Relative paths resolve against the workflow's folder. To keep results, specify export in the workflow. Reruns respect its overwrite/versioning settings. Stopping or failing does not roll back files already written. Wait for completion or stop the automation before closing CurveMole.

Python plugins referenced by a desktop automation must first be reviewed and enabled in **File > Plugin Manager**. Their source fingerprints are checked again at launch; a workflow does not implicitly approve executable plugins. This control runs existing workflows; it is not a visual workflow editor.

16.2 Complete example

```
schema_version: 1
name: Gaussian workflow example
series_name: Synthetic spectra

imports:
  - path: gaussian.csv
    config:
      delimiter: ","
      decimal: "."
      header: true
    columns:
      x: x
      y: [y]
      sigma_y: sigma_y
    aliases: [Synthetic Gaussian]

models:
  - curve: Synthetic Gaussian
    components:
```

```

- function: constant
  name: Background
  parameters:
    offset: 0.0
- function: gaussian
  name: Main band
  parameters:
    area: 3.0
    center: 0.3
    sigma: 1.0
  constraints:
    sigma:
      minimum: 0.001
      maximum: 5.0

fit:
  mode: independent
  curves: [Synthetic Gaussian]
  settings:
    solver: local
    loss: linear
    seed: 1729

export:
  project: gaussian_result.fitproj
  bundle: gaussian_export
  versioned: false
  overwrite: false

```

16.3 Custom formulas in YAML

```

custom_functions:
- identifier: exponential_decay
  display_name: Exponential decay
  kind: generic
  formula: offset + amplitude * exp(-x / tau)
  defaults:
    offset: 0.0
    amplitude: 1.0
    tau: 1.0
  bounds:
    tau: [0.000001, 1000000.0]
  derived:
    lifetime: tau

```

Use the custom identifier in a model component exactly like a built-in function.

16.4 Trusted plugins

List local plugin manifest paths under plugins. Unattended execution also requires one `--trust-plugin IDENTIFIER` argument for each plugin whose Python source may run:

```

curvemole run workflow.yml \
  --trust-plugin org.example.lineshape

```

Trust is never inferred merely because a manifest is present.

17. Python API

17.1 Stable package-level objects

The package root exports GUI-independent classes including Curve, Series, Dataset, Component, Model, Parameter, Project, FitSettings, FitPlan, FitResult, Fitter, FunctionRegistry, and default_registry.

17.2 Minimal fit

```
import numpy as np

from curvemole import Component, Curve, FitSettings, Fitter, Model

x = np.linspace(-5.0, 5.0, 1001)
y = 0.1 + 3.0 * np.exp(-0.5 * ((x - 0.4) / 0.8) ** 2)

curve = Curve("Synthetic band", x, y)
model = Model("One Gaussian")
model.add(Component.create("constant", initial={"offset": 0.1}))
model.add(
    Component.create(
        "gaussian",
        initial={"area": 6.0, "center": 0.3, "sigma": 1.0},
    )
)

result = Fitter().fit_single(curve, model, FitSettings(loss="linear"))
print(result.success)
print(result.statistics)
```

The Gaussian expression used to generate y above is height-parameterized, while the CurveMole Gaussian component is area-parameterized. This is why the initial model area is not simply the synthetic height.

17.3 Bounds and fixed values

```
peak = model.components[1]
peak.parameters["center"].minimum = -1.0
peak.parameters["center"].maximum = 1.0
peak.parameters["sigma"].minimum = 0.05
peak.parameters["sigma"].maximum = 3.0
peak.parameters["area"].fixed = False

for parameter in peak.parameters.values():
    parameter.validate()
```

17.4 Cross-curve links

```
source_path = model_a.parameter_path(curve_a.id, peak_a.id, "center")
peak_b.parameters["center"].link = "${" + source_path + "} + 5.0"
```

Construct a Global FitPlan containing both curve IDs. CurveMole validates the complete graph before optimization.

17.5 Global fit

```
from curvemole.core.fitting import FitMode

plan = FitPlan(
    [curve_a.id, curve_b.id],
    mode=FitMode.GLOBAL,
    settings=FitSettings(loss="linear"),
    spectrum_weights={curve_a.id: 1.0, curve_b.id: 1.0},
    equal_contribution=True,
)

result = Fitter().fit(
    plan,
    {curve_a.id: curve_a, curve_b.id: curve_b},
    {curve_a.id: model_a, curve_b.id: model_b},
)
```

17.6 API compatibility during Preview

The package root is intentionally smaller and more stable than internal modules, but version 0.x may still introduce incompatible changes. Pin the exact CurveMole version in automated research workflows and record it with results.

18. Function Builder and plugins

18.1 Formula Builder

Open **Tools > Function Builder**. Enter:

- an identifier using letters, numbers, and underscores;
- a display name;
- classification as **Peak-shaped function** or **General function**;
- a formula in x ;
- optional formulas for derived area and FWHM;
- for peak functions, optional semantic roles for detected parameters.

Background is assigned to components when adding or editing the model; it is not a separate Function Builder classification. **Use manual-point initialization by default** controls the preferred insertion mode for the new function.

CurveMole detects every non-function symbol other than x as a parameter. Newly detected parameters default to 1.0 and are unbounded. After adding the function to the project library, add a component and edit its initial values and bounds in the normal parameter table.

Example:

```
offset + amplitude * exp(-x / tau)
```

Detected parameters are amplitude, offset, and tau.

For a function classified as a peak, the **Peak parameter roles** table can assign at most one of each supported meaning:

- peak position / centre;
- peak height or peak area / integral;
- FWHM, Gaussian sigma, or HWHM/gamma.

These roles tell graphical placement and automatic peak detection how to initialize parameters even when their names are arbitrary. Height and area roles are mutually exclusive. A width role also supplies the standard derived FWHM automatically when a separate FWHM formula was not entered. Leave genuinely unrelated parameters as **Not specified**. An explicitly unassigned role is respected; CurveMole does not guess a meaning from its name.

18.2 Custom function persistence

The first time a formula is added, CurveMole asks the user to create a dedicated `my_curvemole_functions` folder at a chosen location or open an existing folder with that exact name. Each definition is saved there as `<identifier>.curvemole-function.json`, registered in the current application, and also embedded in the project. Formula, classification, derived quantities, and peak roles travel together.

CurveMole remembers the selected folder and loads every valid function from it at each startup. Reusable functions therefore appear in the normal Add dialog, Quick Add selector, and automatic peak-shape selector without requiring a particular project to be open. Copy the folder between computers to move a personal function library; open that folder when CurveMole asks for the library location.

Project-contained definitions remain portable. When a project reopens, CurveMole reconstructs and validates each restricted expression. A malformed user function or library file is skipped and recorded in the Log rather than executed. Formula files use the safe expression system and are distinct from executable Python plugins.

18.3 Local plugin structure

A local plugin consists of a manifest ending in `.curvemole-plugin.json` and a Python module in the same directory. Example manifest:

```
{
  "identifier": "org.example.my-lineshape",
  "version": "1.0.0",
  "api_compatibility": "1",
  "licence": "GPL-3.0-or-later",
  "capabilities": ["functions"],
  "module": "my_lineshape.py"
}
```

The module exposes `register(api)`. Use `api.register(FunctionDefinition(...))` for model functions or `api.add(kind, id, label, callback)` for exporters, importers, fit algorithms, transformations, analyses, commands, workflows, panels, plot layers and lifecycle notifications. Full contracts and runnable examples are in [Creating and managing plugins](#). Built-ins cannot be replaced through this API; contributed entries carry a plugin-specific symbol and identify the providing plugin in their tooltip.

18.4 Trust boundary

Formula Builder expressions are interpreted safely. Plugins are different: a plugin is Python code and can perform any action allowed to the current user account.

CurveMole reads manifests before executing a module and requires an explicit trust decision. Review source, origin, license, and version before trusting. Do not trust a plugin merely because its identifier resembles a known package.

18.5 Plugin Manager

Open **File > Plugin Manager**, choose a permanent directory, and scan. Review a candidate, then explicitly trust and load it. Enabled plugins load again on restart, independently of projects. **Disable** unloads contributions; **Remove** also forgets the installation. Source files and project plugin data are retained. Remove any components using a plugin before unloading it. Close the manager to refresh menus.

Changed local source needs approval again. Callback errors disable the plugin; failed registration rolls back additions. After a crash or forced termination, CurveMole disables persisted plugins before importing their code and explains recovery. Normal closure clears the session marker. The integrated validated-plugin browser can download selected plugins through **Install selected**. A **Plugins** status badge checks at startup and hourly for updates to loaded supported plugins, lets you select updates, and indicates when a restart is needed to activate staged code. Open **Plugin updates...** in the manager or click the badge, then choose **Update selected**. **Help > Check for updates** checks both CurveMole and loaded plugins. Checking for an update does not install it automatically. Disabled plugins are not checked; local-only plugins without a catalog entry need manual updates. Manual ZIP download remains an alternative. Set `CURVEMOLE_DISABLE_PLUGINS=1` to skip automatic loading manually. This is failure recovery, not a sandbox for Python code.

18.6 Creating extensions beyond model functions

Use `api.add("exporters", "csv", "Laboratory CSV", export_csv)` to add an exporter. Its callback receives the chosen destination in `context.path` and a project snapshot in `context.project`. Built-in export commands remain available. `fit_solvers` callbacks receive constrained residuals, Jacobian, initial values, bounds, settings and cancellation. CurveMole computes standard statistics from the returned result without a second built-in optimization.

Importers, transformations, actions and workflows commit snapshot edits as one undoable operation on success. Other callbacks observe snapshots. JSON data in `context.data` is scoped to the plugin and saved with the project. The [complete plugin author guide](#) provides manifests, exporter and solver code, all callback contracts, limitations and testing instructions.

For a complete exporter, set the manifest's module to `my_exporter.py` and its capabilities to `["exporters"]`, then create this module beside the manifest:

```
import numpy as np

def export_csv(context):
    if context.active_curve_id is None:
        return "Select a spectrum first."
    curve = context.project.dataset.curve(context.active_curve_id)
    np.savetxt(
        context.path,
        np.column_stack([curve.x, curve.y]),
        delimiter=",", header="x,y", comments="",
    )
    return f"Exported {curve.name}"

def register(api):
    api.add("exporters", "csv", "Laboratory CSV", export_csv)
```

After loading this plugin, choose **File > Plugins: Exporters > Laboratory CSV** (marked with the plugin symbol). CurveMole asks for a destination and then runs the callback. The integrated export commands stay available. The module uses NumPy, already included with CurveMole; no additional dependencies are required.

18.7 Community plugin submissions and validation

To share a plugin, open **File > Plugin manager > Share my plugin on GitHub...** This opens the submission guide: fork the repository, upload your complete plugin folder at `custom_plugins/by_community/your_plugin/`, then open a pull request to main. For private use, **Choose plugin folder...** installs it on your computer only; loading a plugin does not publish its source. **Browse validated plugins** opens the downloads that have passed the checks.

The repository's [custom_plugins folder](#) contains the submission guide and downloadable plugins, including a TSV exporter, Fityk project importer, cosmic-ray removal, x-axis calibration and ruby fluorescence pressure monitor. These are separately installed extensions, not automatically enabled CurveMole features. Authors submit a folder with manifest, module, licence, README and functional tests through a pull request. GitHub checks manifests, actual registered capabilities, loading/removal and tests on Linux, Windows and macOS. Only a successful main run publishes the validated catalog and tested plugin ZIP files. **Browse validated plugins** and **Install selected** provide the normal installation route inside the manager, with an explicit trust decision. Alternatively, use the stable

downloads in the [plugin guide](#) or the separate [rolling community plugin release](#), extract the chosen plugin and review/load its manifest in the manager. These direct downloads do not require a GitHub account and are separate from application releases. A source folder or an open PR alone does not indicate validation. Tests demonstrate tested behavior, not complete safety; Python plugins still run with the user's privileges. See the submission guide for required branch protection settings and the limits of validation.

18.8 Acquisition processors and interactive plugin panels

Plugins can register `import_processors` for Automatic folder import (Section 6.4). A processor receives only the detached new acquisition and its settings, runs in a background thread, and returns spectra/models/results to be added as one Undo operation. Enabling a plugin does not start acquisition monitoring.

Plugin panels join the shared dock/tab workspace and can be reopened without losing their current widget state. Supported panel services offer project snapshots, persistent plugin settings and control over the owner's import monitor. A panel may open automatically when its plugin is enabled; monitoring still requires an explicit start. Closing a monitor panel does not stop acquisition. Read each plugin's own README for its scientific assumptions, controls and dependencies.

19. Troubleshooting

19.1 The application does not start on Linux

Run the AppImage from a terminal to capture the exact error. If Qt reports missing EGL, OpenGL, XKB, or XCB libraries, install the distribution packages listed in Section 3.3. If FUSE mounting fails, use `APPIMAGE_EXTRACT_AND_RUN=1`.

The AppImage is built on Ubuntu 22.04 for broad glibc compatibility. Very old Linux distributions may still require a newer runtime.

19.2 Quick Guide, manual, updates, or issue links do not open

CurveMole attempts to launch the host operating system's opener without exposing the AppImage's private Qt and C++ libraries to it. If the opener still fails, the dialog shows the path or URL to copy manually.

Use these direct addresses:

```
https://github.com/SebRoLENS/curvemole
https://github.com/SebRoLENS/curvemole/releases
https://github.com/SebRoLENS/curvemole/issues
```

19.3 Windows SmartScreen or macOS Gatekeeper warns

The Preview executables are not yet commercially signed or notarized. Confirm that the file came from the official release, verify its SHA-256 checksum, and use the operating system's documented override only if you accept the risk.

19.4 Import produces one column

The selected delimiter is probably wrong. Reopen import and compare whitespace, comma, semicolon, tab, or pipe. A decimal comma file commonly uses semicolon as its delimiter. Confirm the header choice and preview before mapping.

19.5 Imported points disappear from fitting

Inspect the Worksheet. A point is excluded when x or y is not finite, σ_y is invalid, a weight is invalid, or the point is masked. Faded markers indicate masks. Use Unmask or undo the transformation that generated missing values.

19.6 A peak appears at the wrong height after placement

Built-in peaks use integrated area, not height. The graphical initializer estimates height relative to the existing model and converts it to area. For overlapping peaks or a poor background, adjust the center target, width handle, and area value before fitting.

19.7 A spline looks unstable

Use fewer, well-spaced nodes. Place nodes where background information exists rather than at every data feature. Graphically placed node y values are fixed by default; unlock only those that the data can independently determine. Nodes created through the API are ordinary free parameters unless explicitly fixed.

19.8 Differential Evolution searches too narrow an interval

Increase the automatic below/above percentages or set physically justified parameter bounds directly. Existing finite bounds override the automatic limits. A linked parameter's resulting value must still satisfy its own bounds.

19.9 The fit is underdetermined or covariance is unavailable

Reduce the number of free parameters, fix known values, add justified links, widen the data range, or improve signal information. More optimizer evaluations cannot create information absent from the data.

19.10 A parameter remains at a bound

First verify the initial value and physical bound. Then inspect correlations and profile likelihood. A bound-active estimate often has an asymmetric uncertainty and may indicate that the data do not locate the optimum within the allowed region.

19.11 A sequential fit pauses

The affected curve becomes active and the status bar displays **Continue sequential fit**. Inspect its propagated model, initial/fitted values, masks, and constraints. The Log and notification identify whether fitting failed, normalized residuals worsened, or a monitored parameter changed beyond its threshold. Correct or approve the model, then continue; this spectrum becomes the new source. For a series with legitimate abrupt changes, adjust or disable the relevant safeguard, or exclude a known discontinuous function from only the parameter-change monitor.

19.12 A project opens read-only

Another instance may hold the sibling `.lock` file. Close the other instance, or use Save As. Remove a lock manually only after confirming it is stale.

19.13 Export refuses to overwrite

Choose a new directory or a versioned export. If updating an existing bundle, keep the project that records ownership of its exported files and explicitly enable update. Older bundles may instead have a `.curvemole-export.json` ownership manifest; retain it. CurveMole will not claim ownership of unrelated colliding files.

19.14 A resampling analysis has many failed replicates

The baseline may be near bounds, poorly initialized, or structurally unstable under perturbation. Increase robustness of the model before increasing replicate count. Report completed and failed counts with any interval.

19.15 An update is detected but cannot be installed automatically

Automatic updating is supported by Linux x86-64 AppImages and Windows x86-64 desktop packages. AppImage and portable replacement requires a writable directory; installed Windows applications use the verified installer. Finish or cancel active tasks. Python/source installations, macOS packages, unsupported architectures, and protected application directories must be updated from the GitHub release page. On Windows, a failed staged update records a helper log in the system temporary directory; include that log when reporting a reproducible updater failure.

19.16 Reporting a problem

Use [GitHub Issues](#). Include:

- CurveMole version;
- operating system and package type;
- exact steps to reproduce;
- expected and observed behavior;
- terminal output or traceback;
- a minimal non-confidential data file if possible.

Review every attachment. Diagnostic material must not contain unpublished spectra, personal information, credentials, or proprietary data unless you intentionally make them public.

20. Reproducible research, privacy, and citation

20.1 Recommended research record

Archive at least:

- exact CurveMole version and DOI;
- original data and import mapping;
- complete `.fitproj`;
- analysis bundle;
- model functions and component order;
- initial values, bounds, fixed values, and links;
- masks and transformations;
- point uncertainties or weights;
- fit mode, spectrum weights, solver, and loss;
- random seed and replicate counts for uncertainty analyses;
- warnings, failed replicates, and residual diagnostics.

20.1.1 Laboratory notebook

The **Laboratory notebook** button at the end of the toolbar opens project notes and a searchable description browser as a workspace tab. Use **Add description** from the context menu of a series, individual spectrum, or added fit function to create or edit its description. Function descriptions are specific to the spectrum containing that function, including when sequential propagation preserves a component ID.

Descriptions are grouped by series and spectrum. Names follow renaming and moves. When an object is removed, its description remains in **Deleted items** with the last known name and an explicit deleted status. Restoring the same object with Undo restores its live status. Copying a function does not duplicate its notes.

Edits in the notebook are kept in the current project as you type; save the `.fitproj` file to persist them. Notebook contents are also included in portable project copies. **Save notebook as text** writes the complete notebook as UTF-8. Alternatively, select **Laboratory notebook (TXT)** in **Export analysis bundle**; this writes `laboratory_notebook.txt` and follows the usual export ownership rules. The notebook can be viewed and exported from a read-only project. Editing is unavailable while a background task runs; a paused sequence allows editing normally.

Series headers are bold and blue, with colors adapted to light and dark themes. A note icon beside a series, spectrum, or fit function opens its description directly. Clearing the description removes its icon and notebook entry.

20.2 Local processing and privacy

CurveMole performs fitting locally. It has no telemetry and does not upload data, projects, models, results, or formulas. Its automatic startup and hourly version check requests public latest-release metadata from GitHub. When plugins are loaded, the application also checks the public validated-plugin catalog at startup and hourly. These checks do not upload scientific data. Opening documentation or release notes, downloading an update, reporting an issue, or using Zenodo is otherwise an explicit external action.

Plugins run locally with the user's permissions and can violate these expectations; trust only reviewed plugin code.

20.3 Citation

If CurveMole contributes to published work, cite the exact version used. The release DOI is inserted into CITATION.cff and the repository README after Zenodo archival. Until archival completes, the versioned GitHub release is the authoritative record:

Romi, S. (2026). *CurveMole: Modular Scientific Curve Fitting* (Version 0.34.0)
[Computer software]. GitHub. <https://github.com/SebRoLENS/curvemole/releases/tag/v0.34.0>

The repository provides **Cite this repository** from CITATION.cff.

20.4 Author and contact

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20.5 License

CurveMole is released under GPL-3.0-or-later. User data, private projects, results, and unpublished formulas remain under the user's control. Publicly distributed extensions must follow their applicable license obligations.

Appendix A. Keyboard shortcuts

Shortcuts use the platform's standard key sequence where applicable.

Shortcut	Action
Ctrl+N	New project
Ctrl+O	Open project
Ctrl+I	Import data
Ctrl+S	Save project
Ctrl+Shift+S	Save project as on common desktop mappings
Ctrl+E	Export analysis bundle
Ctrl+Shift+E	Export one numeric file per selected spectrum
Ctrl+Z	Undo
Standard Redo shortcut	Redo
Ctrl++	Add component
F5	Open Fit dialog
F1 or platform Help shortcut	Quick Start
Esc	Cancel graphical peak or manual-point placement
Ctrl while dragging a normal peak handle	Change a fixed graphical parameter without unfixing it
Double-click during manual-point placement	Finish after the minimum number of points
Right-click during manual-point placement	Remove the nearest point
Right-drag / left-drag with Mask active	Mask / unmask an interval
Up / Down with plot focus	Activate previous / next spectrum

Appendix B. Built-in identifiers

Identifier	Display name	Parameters
gaussian	Gaussian	area, center, sigma
lorentzian	Lorentzian	area, center, gamma
voigt	Voigt	area, center, sigma, gamma
pseudo_voigt	Pseudo-Voigt	area, center, fwhm, eta
constant	Constant	offset
linear	Linear	intercept, slope
polynomial	Polynomial	c0 through selected order
cubic_spline	Cubic spline	y0 through last node

Appendix C. File extensions

Extension	Purpose
.fitproj	Complete CurveMole project
.fitmodel	Reusable model and formula definitions
.curvemole-function.json	Reusable safe formula in my_curvemole_functions
.curvemole-plugin.json	Manifest for an executable Python plugin
.yml, .yaml	Reproducible workflow
Any text-file suffix	Imported data when the contents form a valid numeric table
.whl	Installable Python package
.AppImage	Linux desktop package
.exe	Windows installer or portable executable
.zip	Windows portable package or plugin download
.dmg	macOS disk image

Appendix D. Preview 0.34.0 limitations

The following boundaries are important when evaluating this release:

- scientific validation toward 1.0.0 is still in progress;
- local optimizers include nonlinear least squares, Nelder-Mead, Powell, and L-BFGS-B; none guarantees the global optimum for an arbitrary nonlinear model;
- `sigma_x` is stored but not used in optimization;
- graphical links support a source-parameter picker; complex multi-source relations still require the advanced expression editor;
- fit ranges exist in the core model but do not yet have a complete graphical editor;
- the GUI does not yet provide one consolidated table for every fit statistic;
- Windows and macOS packages may be unsigned; Linux release provenance is available through GitHub artifact attestation;
- the interface and manual are currently maintained in English;
- plugin API and other 0.x APIs may change before 1.0.0.

These limitations are stated so that testing can focus on genuine behavior rather than inferred promises. Report unclear workflows and reproducible defects through the issue tracker.

Appendix E. Building this manual

The Markdown file you are reading is the only hand-edited manual source. From a source checkout with Pandoc, XeLaTeX, and latexmk installed, run:

```
python scripts/build_manual.py
```

The command validates version declarations and local links, then creates:

```
docs/CurveMole_User_Manual.tex  
docs/CurveMole_User_Manual.pdf
```

GitHub Actions repeats this process for documentation changes and releases. Release assets use versioned filenames and include both LaTeX and PDF editions. Do not edit the generated files directly; changes will be replaced by the next automated build.